Bhuvan Arora

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Research Interests

Evaluating and training LLM agents on long-horizon tasks, reinforcement-learning environments built from real-world settings, multilingual and Indic NLP, and cultural bias in large language models.

Education

Birla Institute of Technology and Science (BITS) Pilani Pilani, Rajasthan, India
B.E. Computer Science; CGPA: 9.57/10 *2023 – May 2027 (expected)*
Relevant coursework: Machine Learning, Introduction to LLMs, Design and Analysis of Algorithms, Probability and Statistics, Theory of Computation

Publications and Preprints

KiranaBench: Can a Language Model Run a Neighbourhood Shop?
Bhuvan Arora, Ankita Karmakar, Chaitanya Jain, Dilesh Churiwal, Harshit Jain, Shivansh Manish Dubey, Manish Prasad, Raja Piyush, Kshitij Tayal, Dhruv Kumar, Jagat Sesh Challa, Raunak Manekar, Yash Sinha
Under review at ACL Rolling Review (October 2026 cycle), committing to NAACL 2027. arXiv:2610.22980

DiSCo: A Distribution-First Steering and Cultural Prior Evaluation Framework for Measuring Cultural Preference Bias in LLMs
Bhuvan Arora, Devesh Saraogi, Sravya Varada, Dhruv Kumar
Preprint, 2026. arXiv:2609.10253

LLM-Driven Security and Resilience in 6G Mission-Critical Communication Networks
Siva Sai, **Bhuvan Arora**, Vineet Suri, Vinay Chamola, Mohan Gurusamy
IEEE Open Journal of the Communications Society, vol. 7, pp. 8963–8984, 2026 (survey).
doi:10.1109/OJCOMS.2026.3717648

Research Experience

Karobaar: Indian business simulations as RL environments for LLM agents 2026 – Present
Project lead; Advisor: Dr. Dhruv Kumar *BITS Pilani, India*

- Leading a team of student researchers building simulated Indian businesses (kirana shop, truck fleet, municipal corporation, wedding planning, staffing agency, scrap dealer, hotel and cloud kitchen) in which an LLM agent runs the business over a long horizon and is scored against the simulator's ground truth.
- First environment, **KiranaBench** (first author, under review): one model runs every job in a neighbourhood shop for 30 days, dealing with customers and suppliers in English, Hindi and Hinglish. Four open-weight models fall well short of a simple rule-based shopkeeper and write worse to Hindi and Hinglish suppliers than to English ones.
- Turning these simulations into reinforcement-learning environments for training LLM agents.

BharatGUI: computer-use agents on Indian government and public-service websites 2026 – Present
Advisor: Dr. Dhruv Kumar *BITS Pilani, India*

- Building realistic, self-hosted replicas of Indian websites, in English and Hindi, as environments for testing and training screenshot-based computer-use agents, with every task checked against the site's own database.
- Built so far: a GST tax-filing portal (191 pages, 117 tasks) and an IRCTC-style railway-booking portal (109 tasks), each with an automatic grader and adjustable rendering quality; more Indian portals are planned.

DiSCo: Measuring cultural preference bias in LLMs 2026
First author; Advisor: Dr. Dhruv Kumar *BITS Pilani, India*

- Developed a distribution-first evaluation framework for how LLMs handle culturally specific norms and values, beyond language differences alone.

Physical AI companion for older adults Ongoing

Advisor: Dr. Siddharth Mehrotra *BITS Pilani, India*

- Built a working prototype of a voice assistant operated with physical buttons and knobs, designed for older adults with limited digital literacy. It supports open-ended conversation, web search, and reminders and alarms.
- Building an internet-based calling, voice-note and messaging backend that connects users to family members through a companion app.

Industry Experience

Samsung R&D Institute India (SRI-B) May 2026 – Jul 2026

Advanced Developer Intern (MAGPIE); received a pre-placement offer *Bengaluru, India*

- **AI planner app:** conversational daily planner using Google's A2UI (Agent-to-UI) protocol for input and confirmation, with location-aware clash detection (Nominatim, OpenRouteService). Tested whether A2UI, designed for remote agents, can run on-device, with partial success.
- **Damaged-barcode recovery:** two-stage retrieval pipeline (FFT-based embeddings with approximate nearest-neighbour shortlisting, then cross-correlation reranking) that reached **95.75% accuracy across 9 barcode types**.

CloudDefense.AI May 2025 – Jul 2025

Practice School I Intern

Other Projects

- **local-flow** (2026): open-source dictation tool that runs fully locally. GitHub
- **AI Buddies Multi-Agent System** (2025): multi-agent LLM application. GitHub
- **BITS Connect** (2025): Flutter and Firebase app that lets BITS Pilani students form study groups and meetups. GitHub

Skills and Certifications

- **Research:** LLM agents; simulation-based benchmarks and RL environments; computer-use agents; multilingual and cultural evaluation
- **Tools:** Python; vLLM, Hugging Face Transformers, PEFT, Ollama; ElevenLabs voice stack; Flutter, Firebase
- **Certifications:** NVIDIA DLI Fundamentals of Deep Learning (2025)